Unsupervised Hyperspectral Band Selection With Multigraph Integrated Embedding and Robust Self-Contained Regression

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***Abstract*— Band selection is an effective means to alleviate the curse of dimensionality in hyperspectral data. Many methods select a compact and low redundant band subset, which is inadequate as it may degrade the classification performance. Instead, more emphasis shall be put on selecting representative bands. In this article, we propose a robust unsupervised band selection method to address this issue. Our method reveals bandwise representativeness based on the comprehensive interband neighborhood structure. It incorporates an interband neighborhood graph into a sparse self-contained regression model in order to provide a reasonable measure for bandwise representativeness. The derived coefficient matrix not only uncovers bandwise importance values but also is coherent to the generalized interband local neighborhood structure. For constructing the interband neighboring structural graph, an integrated multigraph model is employed to achieve better generalization performance. It combines the benefit of multiple graphs but is insusceptible to the defects of a single one. To enhance the reliability of this model, a joint trace minimum and nonnegative constraint is imposed on the coefficient matrix. Accordingly, a multigraph integrated embedding and robust self-contained regression model (MGRSR) is formulated. In addition, an iterative update algorithm is developed to solve the problem. Comparative experiments on three hyperspectral data sets illustrate that MGRSR is robust to various data and has superior performance compared with several state-of-the-art methods.**

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I. Introduction

HYPERSPECTRAL imaging acquires hundreds of narrow and contiguous spectral bands of ground targets [1]–[3]. Benefiting from the abundant spectral information, hyperspectral images (HSIs) are of great potential in image classification, object detection, and scene understanding [4]–[8]. Unfortunately, due to limited labeled samples, excessive spectral bands easily incur deteriorated classification performance due to the curse of dimensionality [9], [10]. Furthermore, correlated, irrelevant, and noisy bands also tremendously increase the cost of data analysis and storage [11]. In this regard, dimensionality reduction is an important task in hyperspectral data analysis [12], [13].

Dimensionality reduction techniques can be generally divided into two categories: feature extraction and band selection [11]. Feature extraction establishes appropriate transformation to map the data to low dimensional subspaces [14]. Classic methods include principal component analysis (PCA) [15], locally linear embedding (LLE) [16], locality-preserving projections (LPP) [17], and so on. While these methods work well in data dimension reduction, the physical meaning of the spectrum is severely distorted. As a result, this exerts a strong adverse impact on the interpretability of the materials [18]. On the contrary, band selection is designed to select appropriate bands without distorting the spectral responses of objects, which is the focus of this article.

According to whether labeled samples are involved, existing band selection methods can be grouped into supervised [19], [20], semisupervised [21]–[24], and unsupervised methods [25]–[27]. With the guidance of label information, supervised and semisupervised methods are advantageous in achieving sound classification or recognition performance [28]. Nevertheless, producing labels for hyperspectral remote sensing data is expensive, time-consuming, and tedious. From this perspective, unsupervised band selection does not require labeled data, offering a feasible solution to many remote sensing applications.

Current unsupervised band selection methods can be divided into ranking-, clustering-, searching- [11], sparsity- [29], and

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graph embedding-based methods. Ranking-based methods are conducted to gather the top-ranked bands in accordance with predefined metrics [30], [31]. For example, in linearly constrained minimum variance-based band correlation constraint (LCMV-BCC) [32], correlation-related criteria are applied to attain desired band subset. The same applies for linearly constrained minimum variance-based band correlation minimization (LCMV-BCM) [32]. The joint Ward's linkage-based hierarchical clustering and divergence-based prioritization (WaLuDi) [33] is a typical clustering-based method, which considers both interband correlation and bandwise necessity. Other clustering approaches include enhanced fast density-peak-based clustering (E-FDPC) [34] and optimal clustering with top-rank cut (TRC-OC-FDPC) [35]. The research found that many clustering-based methods are adept at reducing redundancy, while the overall usefulness of the selected bands for a specific task is usually ignored.

The searching-based methods focus on finding the subset with optimal overall performance [36], [37]. For example, Gong *et al.* [38] constructed the objective functions with a conflicting relationship between total information entropy and the number of selected bands. For volume-gradient-based band selection (VGBS) [39], the core goal is to minimize the redundancy based on the gradient of the constructed parallelotope volume. In addition, to collect the informative bands, measures associated with the gradient information are investigated in the dominant set extraction-based band selector (DSEBS) [40]. In terms of low-noise HSIs, it is undoubted that bandwise gradient information can reflect its information richness. This enables DSEBS to acquire valuable bands conducive to classification.

Sparsity-based methods build a small band subset in the light of sparse representation theory. Taking sparse representation-based band selector (SpaBS) [29] as an example, its occurrence frequency of coefficient matrix with sparse representation is generated to characterize the bandwise prioritization. Essentially, SpaBS is favorable to eliminating the correlated bands to some extent. Except for the high computational complexity, SpaBS is highly dependent on the learned dictionary, which seriously influences its stability and reliability.

The graph embedding-based methods aim to find a few bands capable of retaining the intrinsic geometric structure information, i.e., manifold, of the original data [41]. A typical method is the Laplacian Eigenmaps [42]. This method attempts to find an appropriate map, which preserves the local neighborhood structure similar to high-dimensional space, while, just with the orthogonality constraint, the obtained map may contain negative values, which is the noise information and disadvantageous to form a reasonable band selection criterion. Regarding this, the robust principal component analysis (RPCA) [43] assumes that the data can be divided into the low-rank and sparse parts. Specifically, the low-rank part corresponds to the essential components, while the sparse part represents the noise components. Meanwhile, in the sparse and low-rank graph for discriminant analysis (SLGDA) [44], the low-rank and sparsity constraints are introduced into the self-contained regression model, while, in self-contained regression [25], [45], there exists a difficulty, that is, the solved coefficient matrix may be invalid.

Moreover, in the graph embedding-based methods, the noise and outliers of the raw data can result in an unreliable graph. Then, to formulate an accurate and robust graph, the multiview graph on basis of the multiple-view data is investigated [46]–[48]. In terms of the latent multiview semisupervised classification (LMSSC) [49], it proposes to construct the multigraph based on the learned latent representation rather than the original multiview data. Despite the promising performance of these methods, multiview data can have a high acquisition cost. Considering that different definitions of neighborhood graphs have their own merits, it is sensible to comprehensively fuse them to obtain an integrated graph [50]. Note that, to achieve satisfactory band selection results, the graph should also embody the intrinsic structural information of the data.

In this article, we propose a robust unsupervised band selection method, which enforces bandwise representativeness to be consistent with the comprehensive interband neighborhood relations. In specific, to construct a reliable measure for bandwise representativeness, interband neighborhood graphs are incorporated into a sparse self-contained regression model. This improves the regression model not only by enabling the regression coefficient matrix to reflect bandwise importance values but also making it consistent with the interband local geometric structure. Note that multiple graphs can be built to reflect different bandwise relationships. To provide a synthetical characterization of interband neighborhood graphs, an integrated multigraph is utilized to combine the merits of multiple graphs. This leads to better generalization performance of the proposed model. Moreover, to enhance the reliability of the model and circumvent trivial solutions, a joint trace minimum and nonnegative constraint is imposed on the regression coefficient matrix. Consequently, a multigraph integrated embedding and robust self-contained regression model (MGRSR) is formulated. We further develop an iterative update algorithm based on the alternating direction method of multipliers (ADMM) [51], [52] to optimize the MGRSR model. To evaluate the performance of the proposed method, comparative experiments with several state-of-the-art methods have been conducted on three popular hyperspectral data sets. Experimental results manifest the robustness and effectiveness of the proposed method.

The main contributions of this article are summarized as follows.

1) This article presents a robust unsupervised band selection method with MGRSR. The interband neighborhood graph is incorporated into the sparse self-regression model. In this regard, the regression coefficients are not only able to embody bandwise contribution in constituting the original data but also conformed to the interband manifold structure. Apparently, it is beneficial to gain reliable bandwise representativeness measures favorable to classification. To give a comprehensive description of the neighborhood graph between bands, the integrated multigraph from different perspectives is used. This helps to remedy the deficiency of a single graph and

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TABLE I
NOTATIONS

N	Number of pixels
d	Number of spectral bands
n	Number of iterations
$\boldsymbol{X} \in \mathbb{R}^{N\times d}$	Hyperspectral image
$\boldsymbol{x}_i \in \mathbb{R}^{N\times 1}$	The ith band in $\boldsymbol{X}$
$\mathbf{A} \in \mathbb{R}^{\mathbf{d}\times \mathbf{d}}$	Coefficient matrix
$\boldsymbol{a}^i \in \mathbb{R}^{1\times d}$	The ith row of $\boldsymbol{A}$
$\boldsymbol{a}_j \in \mathbb{R}^{d\times 1}$	The jth column of $\boldsymbol{A}$
$\boldsymbol{Y} \in \mathbb{R}^{N\times d}$	Dictionary matrix
$\boldsymbol{E} \in \mathbb{R}^{N\times d}$	Residual matrix
$\boldsymbol{I} \in \mathbb{R}^{d\times d}$	Identity matrix
$\boldsymbol{S} \in \mathbb{R}^{d\times d}$	Inter-band neighborhood graph
$N(\boldsymbol{x}_i)$	The k nearest neighbors of $\boldsymbol{x}_i$
$\boldsymbol{M} \in \mathbb{R}^{N\times d}$	Estimated residual matrix
$\boldsymbol{F} \in \mathbb{R}^{d\times d}$	Integrated inter-band neighborhood graph
$\boldsymbol{G} \in \mathbb{R}^{d\times d}$	$\boldsymbol{G} = [g_{ij}]_{ij}$, $g_{ij} = \left\|\boldsymbol{a}^i - \boldsymbol{a}^j\right\|_2^2$
$\boldsymbol{L_s} \in \mathbb{R}^{d\times d}$	Laplacian matrix
$\boldsymbol{D} \in \mathbb{R}^{d\times d}$	Degree matrix
$\boldsymbol{U} \in \mathbb{R}^{d\times d}$	Diagonal matrix
$\epsilon_{ij}, s_{ij}, a_{ij}, m_{ij}$	Elements in the ith row and the jth column of $\boldsymbol{E}$, $\boldsymbol{S}$, $\boldsymbol{A}$, $\boldsymbol{M}$
$\boldsymbol{w} \in \mathbb{R}^{d\times 1}$	Significance vector

attain a comprehensive interband neighborhood graph with better generalization performance.

2) Apart from the row-sparsity constraint, this article proposes to impose a joint trace minimum and nonnegative constraint on the self-regression model. Accordingly, an MGRSR model is established. It is noticeable that the rigorous constraint can prevent the regression coefficient matrix from being a trivial solution, i.e., the identity matrix. Moreover, the situation that bands are constructed by themselves can be effectively suppressed. Hence, the reliability of the formulated model gets enhanced accordingly.
3) To solve the presented problem, this article also develops an iterative update algorithm in line with ADMM. In addition, concerning evaluating the performance of MGRSR, comparative experiments are carried out on three widely adopted hyperspectral data sets. Experimental results demonstrate that, compared with others, the proposed method is robust and exhibits obvious superiority.

The rest of this article is organized as follows. In Section II, we briefly review some related work. Section III gives a detailed description of the proposed band selection method, including the establishment of MGRSR and the corresponding optimization algorithm to tackle MGRSR. Section IV presents the comparative experimental results and discussion. The conclusions and future work are described in Section V.

II. Related Work

In this section, the notations and dimensions of the main variables used in this article are first listed through Table I. Then, we give a brief review of the sparse representation and self-contained regression algorithms related to our proposed method.

A. Sparse Representation

Sparse representation has been widely used for HSI analysis [25], [53], [54]. The primary assumption of sparse representation is that the original data can be approximated by the product of a coefficient matrix with few nonzero elements and a dictionary [29], [55]. Assume that $\boldsymbol{X}$ is an HSI with N pixels and d bands. Then, the sparse representation problem of $\boldsymbol{X}$ can be expressed as

$$\begin{aligned} &\min \|\boldsymbol{E}\|_p + \gamma_1 \|\boldsymbol{A}\|_q \\ &\text{s.t. } \boldsymbol{X} = \boldsymbol{Y}\boldsymbol{A} + \boldsymbol{E} \end{aligned} \tag{1}$$

where $\boldsymbol{E}$ is a residual matrix, where $\boldsymbol{A}$ denotes the coefficient matrix, $\boldsymbol{Y}$ is the dictionary, $\|\boldsymbol{E}\|_p$ represents the l_p-norm of $\boldsymbol{E}$, $\|\boldsymbol{A}\|_q$ means the l_q-norm of $\boldsymbol{A}$, and γ_1 is a regularization parameter used to balance two terms.

In (1), $\|\boldsymbol{E}\|_p$ is used to make the residual as small as possible. Meanwhile, $\|\boldsymbol{A}\|_q$ is helpful to ensure the sparsity of $\boldsymbol{A}$. Generally, p or q can be set to 0, 1, or 2. When $p = 1$, $\|\boldsymbol{E}\|_p$ is the lasso regularization of $\boldsymbol{E}$ and is defined as $\|\boldsymbol{E}\|_1 = \sum_{i=1}^{N}\sum_{j=1}^{d}|\epsilon_{ij}|$, where $|\cdot|$ represents the absolute operator. In this definition, the coefficient matrix is mainly subject to the learned dictionary. Note that dictionary learning is a time-consuming process with many algorithms developed [56], [57].

B. Self-Contained Regression

Compared with the sparse representation technique, self-contained regression [45], [58] is usually more efficient. The reason is that it directly employs the original data as the dictionary, so the complex dictionary learning can be effectively avoided. Given $\boldsymbol{X}$, the self-contained regression model is defined as

$$\begin{aligned} &\min \|\boldsymbol{E}\|_1 \\ &\text{s.t. } \boldsymbol{X} = \boldsymbol{X}\boldsymbol{A} + \boldsymbol{E} \end{aligned} \tag{2}$$

where $\boldsymbol{X}$ on the right-hand side of (2) characterizes the dictionary used to regress the HSI $\boldsymbol{X}$ on the left-hand side.

Concerning the redundancy in an HSI $\boldsymbol{X}$, it is reasonable to use as few columns of the dictionary $\boldsymbol{X}$ as possible to approximate data $\boldsymbol{X}$. Analogous to sparse representation, the sparsity constraint associated with $\boldsymbol{A}$ can also be applied to the self-contained regression model. Specifically, the $l_{2,1}$-norm is frequently adopted to achieve a row-sparsity matrix [58]. With the $l_{2,1}$-norm of $\boldsymbol{A}$, the sparse self-contained regression model can be further reformulated as

$$\begin{aligned} &\min \|\boldsymbol{E}\|_1 + \gamma_1 \|\boldsymbol{A}\|_{2,1} \\ &\text{s.t. } \boldsymbol{X} = \boldsymbol{X}\boldsymbol{A} + \boldsymbol{E} \end{aligned} \tag{3}$$

where γ_1 is a regularization parameter, and $\|\boldsymbol{A}\|_{2,1}$ is defined as $\|\mathbf{A}\|_{2,1} = \sum_{i=1}^{d}\|\boldsymbol{a}^i\|_2 = \sum_{i=1}^{d}(\sum_{j=1}^{d}a_{ij}^2)^{1/2}$. As a regression coefficient matrix, the row vector $\boldsymbol{a}^i$ illustrates the importance of the ith band in constituting $\boldsymbol{X}$. In this regard, self-contained regression can be exploited for hyperspectral band selection.

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III. Proposed Band Selection Method

In this section, we introduced the details of the proposed method for unsupervised hyperspectral band selection with multigraph autofused embedding and robust self-contained regression.

A. Formulation of Multigraph Integrated and Robust Self-Contained Regression (MGRSR)

1) Nonnegative Constraint and Trace Minimization: Given the problem defined in (3), a valid solution is $\boldsymbol{A} = \boldsymbol{I}$, where $\boldsymbol{I}$ is an identity matrix with size $d \times d$. In this solution, each band is directly represented by itself. Consequently, the bandwise contribution is equal but worthless for band selection. Therefore, it is necessary to introduce additional constraints to circumvent this trivial solution.

To address this issue, we propose to impose a joint trace minimization and nonnegative constraint on $\boldsymbol{A}$, so (3) can be reformulated as

$$\begin{aligned} &\min \|\boldsymbol{E}\|_1 + \gamma_1 \|\boldsymbol{A}\|_{2,1} + \alpha \mathrm{Tr}(\boldsymbol{A}) \\ &\text{s.t. } \boldsymbol{A} \geq 0, \\ &\quad \boldsymbol{X} = \boldsymbol{X}\boldsymbol{A} + \boldsymbol{E} \end{aligned} \tag{4}$$

where $\mathrm{Tr}(\cdot)$ represents the trace operator, and α is a tradeoff parameter.

It is well known that the trace of a matrix is the sum of its diagonal elements. In this connection, for $\boldsymbol{A} \geq 0$, the further constraint of trace minimization of $\boldsymbol{A}$ will enforce the diagonal elements of it to be 0. Therefore, the joint constraints of nonnegativity and trace minimization of $\boldsymbol{A}$ can prohibit $\boldsymbol{A}$ from being the identity matrix $\boldsymbol{I}$. Then, the situation that each band is regressed by itself can be effectively inhibited. Benefiting from this, $\boldsymbol{A}$ is capable of providing more reliable bandwise significance.

2) Interband Neighborhood Graph Constraint: Note that correlated bands should possess similar capacity in terms of representativeness or significance. This implies that bandwise significance should be coherent with the interband neighborhood relationship. In turn, it is required that $\boldsymbol{A}$ follows the interband neighborhood relationship rather than being arbitrary, which cannot be achieved by (4). To address this issue, we propose an interband neighborhood graph restraint to enhance the reasonability of $\boldsymbol{A}$, which is defined as

$$\begin{aligned} &\min \|\boldsymbol{E}\|_1 + \gamma_1 \|\boldsymbol{A}\|_{2,1} + \alpha \mathrm{Tr}(\boldsymbol{A}) \\ &\quad + \gamma_2 \sum_{i=1}^{d} \sum_{j=1}^{d} \left\|\boldsymbol{a}^i - \boldsymbol{a}^j\right\|_2^2 s_{ij} \\ &\text{s.t. } \boldsymbol{A} \geq 0, \\ &\quad \boldsymbol{X} = \boldsymbol{X}\boldsymbol{A} + \boldsymbol{E} \end{aligned} \tag{5}$$

where γ_2 is a balance parameter, and $s_{i,j}$ from $\boldsymbol{S}$ is capable of characterizing the neighborhood relationship between $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$. In this connection, the interband neighborhood graph $\boldsymbol{S}$ plays a crucial role in achieving a structure-preserving $\boldsymbol{A}$. Consequently, $\boldsymbol{A}$ is not only able to reflect the bandwise value but also in accordance to the interband local neighborhood structure.

Given various considerations, the interband neighborhood graph $\boldsymbol{S}$ can be defined differently to depict a variety of local structures. Each definition has its advantages and limitations, so it is difficult to determine which one shall be used. A solution to this problem is combining multiple graphs to improve the generalization capability and robustness of the model [48], [59], [60]. In this way, instead of exploring a single interband structural graph, multiple graphs from different perspectives can be integrated to produce a comprehensive solution that combines the advantages of each graph. To this end, this article employs an integrated function from three perspectives to generate an effective interband structure constraint on $\boldsymbol{A}$. Therefore, a multigraph integrated robust self-contained regression (MGRSR) model is formulated as

$$\begin{aligned} &\min \|\boldsymbol{E}\|_1 + \gamma_1 \|\boldsymbol{A}\|_{2,1} + \alpha \mathrm{Tr}(\boldsymbol{A}) \\ &\quad + \gamma_2 \sum_{i=1}^{d} \sum_{j=1}^{d} \left\|\boldsymbol{a}^i - \boldsymbol{a}^j\right\|_2^2 s_{ij} \\ &\quad + \lambda \sum_{q=1}^{3} h_q \left\|f_q(\boldsymbol{X}) - \boldsymbol{S}\right\|_F^2 \\ &\text{s.t. } \boldsymbol{A} \geq 0, \\ &\quad \boldsymbol{S} \geq 0, \\ &\quad \boldsymbol{X} = \boldsymbol{X}\boldsymbol{A} + \boldsymbol{E} \end{aligned} \tag{6}$$

where λ is the regularization parameter, and f_1, f_2, and f_3 represent three broadly utilized adjacent structural functions, respectively. Regarding HSI $\boldsymbol{X}$, $f_1(\boldsymbol{X})$, $f_2(\boldsymbol{X})$, and $f_3(\boldsymbol{X})$ are three interband neighborhood graphs of size $d \times d$, which depicts the neighborhood relationship between all bands from three perspectives. h_q is a weight coefficient used to ensure that the last term is minimized overall.

In specific, if $\|f_q(\boldsymbol{X}) - \boldsymbol{S}\|_F^2$ is large, then, to minimize the last term, h_q should be reduced and vice versa. For simplicity, it can be set as $1/\|f_q(\boldsymbol{X}) - \boldsymbol{S}\|_F$, which is adaptive with no extra parameters required [50]. $\|\cdot\|_F$ represents the Frobenius norm. For example, the Frobenius norm of $\mathbf{A}$ can be expressed as $\|\mathbf{A}\|_F = (\sum_{i=1}^{d} \sum_{j=1}^{d} |a_{ij}|^2)^{1/2}$.

Evidently, from (6), we can find that $\boldsymbol{S}$ is an integrated multigraph rather than a single graph, which helps to realize the complementary advantages of three graphs $f_1(\boldsymbol{X}), f_2(\boldsymbol{X})$, and $f_3(\boldsymbol{X})$. In addition, the generalization capability of the interband neighboring structural graph can be improved through the integration of multiple graphs as well.

Let $f_q(\boldsymbol{x}_i, \boldsymbol{x}_j)$ be the element of ith row and the jth column in $f_q(\boldsymbol{X})$. To obtain effective multigraph, this article adopts the Gaussian kernel function [61], [62], the effective neighborhood function [63], and the spectral angle function [64] to define $f_1(\boldsymbol{x}_i, \boldsymbol{x}_j)$, $f_2(\boldsymbol{x}_i, \boldsymbol{x}_j)$, and $f_3(\boldsymbol{x}_i, \boldsymbol{x}_j)$, respectively. In specific, the Gaussian kernel function for $f_1(\boldsymbol{x}_i, \boldsymbol{x}_j)$ is defined as

$$f_1(\boldsymbol{x}_i, \boldsymbol{x}_j) = \begin{cases} e^{-\frac{\left\|\boldsymbol{x}_i - \boldsymbol{x}_j\right\|_2^2}{\theta}}, & \boldsymbol{x}_j \in N(\boldsymbol{x}_i) \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

where θ is the kernel parameter used to adjust the scale of data. $f_1(\boldsymbol{x}_i, \boldsymbol{x}_j)$ offers the explicit interband neighboring relationship between $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$. If $\boldsymbol{x}_i$ belongs to the k nearest

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neighbors of $\boldsymbol{x}_j$, $f_1(\boldsymbol{x}_i, \boldsymbol{x}_j)$ is computable; otherwise, it is set to 0. For varying scales of data, the neighborhood structure described by the Gaussian kernel function can change by tuning the value of θ. In contrast, the similarity measure given in [63] is scale invariant, which is used to define $f_2(\boldsymbol{x}_i, \boldsymbol{x}_j)$ as

$$f_2(\boldsymbol{x}_i, \boldsymbol{x}_j) = \begin{cases} \dfrac{e_{i,k+1} - e_{ij}}{k e_{i,k+1} - \sum_{t=1}^{d} e_{i,t}}, & \boldsymbol{x}_j \in N(\boldsymbol{x}_i) \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

where $e_{i,k+1}$ represents the $(k+1)$th smallest Euclidean distance between $\boldsymbol{x}_i$ and the other bands in $\boldsymbol{X}$. Meanwhile, the Euclidean distance between the ith band and the jth one is denoted as e_{ij}. It is obvious that, compared with $f_1(\boldsymbol{x}_i, \boldsymbol{x}_j)$, $f_2(\boldsymbol{x}_i, \boldsymbol{x}_j)$ has the advantage of scale invariance. Moreover, no kernel parameter setting is required for (8).

Note that the spectral angle function is also broadly applied to describe adjacent neighborhood [64]. Therefore, we take spectral angle function as $f_3(\boldsymbol{x}_i, \boldsymbol{x}_j)$ to measure the adjacent relationship

$$f_3(\boldsymbol{x}_i, \boldsymbol{x}_j) = \begin{cases} \dfrac{\boldsymbol{x}_i^T \boldsymbol{x}_j^T}{\|\boldsymbol{x}_i\| \|\boldsymbol{x}_j\|}, & \boldsymbol{x}_i \in N(\boldsymbol{x}_j) \text{ or } \boldsymbol{x}_j \in N(\boldsymbol{x}_i) \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

Evidently, based on (7), (8), and (9), $f_q(\boldsymbol{X})(q = 1, 2, 3)$ in (6) can be determined. In this case, the integrated multigraph is capable of comprehensively combining the interband local structural information from different views and attain better generalization capacity. This contributes to acquiring a reliable coefficient matrix $\boldsymbol{A}$, which is able to embody the bandwise importance value, as well as being consistent to the comprehensive interband local neighborhood structure.

B. Optimization of MGRSR

We introduce an iterative update algorithm in this article to provide the solution to (6). The specific procedure to obtain $\boldsymbol{E}$, $\boldsymbol{S}$, and $\boldsymbol{A}$ is described as follows.

1) Fix $\boldsymbol{S}$ and $\boldsymbol{A}$ and Optimize $\boldsymbol{E}$: In (6), after fixing $\boldsymbol{S}$ and $\boldsymbol{A}$, the objective function with respect to $\boldsymbol{E}$ is equivalent to (2). By introducing the Lagrangian multiplier μ_1, (2) can be transformed to an unconstrained optimization problem as follows:

$$\min \|\boldsymbol{E}\|_1 + \mu_1 \|\boldsymbol{X} - \boldsymbol{X}\boldsymbol{A} - \boldsymbol{E}\|_F^2. \tag{10}$$

Assume that $\boldsymbol{M} = \boldsymbol{X} - \boldsymbol{X}\boldsymbol{A}$. Then, after setting the derivative of (10) with respect to ε_{ij} to zero, we have

$$\varepsilon_{ij} = \begin{cases} m_{ij} - \dfrac{1}{2\mu_1}, & \varepsilon_{ij} > 0 \\ m_{ij} + \dfrac{1}{2\mu_1}, & \varepsilon_{ij} < 0 \\ 0, & \varepsilon_{ij} = 0. \end{cases} \tag{11}$$

Then, by employing the sign function, (11) can be simplified to

$$\epsilon_{ij} = \operatorname{sign}(m_{ij}) \max\left(0, \left|m_{ij}\right| - \frac{1}{2\mu_1}\right) \tag{12}$$

where $|m_{ij}|$ depicts the absolute value of m_{ij}, and $\operatorname{sign}(\cdot)$ illustrates the sign function. If $m_{ij} > 0$, $\operatorname{sign}(m_{ij}) = 1$; if $m_{ij} < 0$, $\operatorname{sign}(m_{ij}) = -1$; otherwise, $\operatorname{sign}(m_{ij}) = 0$. Then, based on (12), we can attain the $(n+1)$th explicit iterative rule related to $\boldsymbol{E}$ expressed as

$$\boldsymbol{E}^{(n+1)} = \operatorname{sign}(\boldsymbol{M}) \max\left(0, |\boldsymbol{M}| - \frac{1}{2\mu_1}\right) \tag{13}$$

where n denotes the number of iterations.

Evidently, $\boldsymbol{E}$ can be iteratively updated via (13).

2) Fix $\boldsymbol{E}$ and $\boldsymbol{A}$, Optimize $\boldsymbol{S}$:: After fixing $\boldsymbol{E}$ and $\boldsymbol{A}$, we can find the objective function with respect to $\boldsymbol{S}$ given by

$$\begin{aligned} &\min \gamma_2 \sum_i \sum_j \left\|\boldsymbol{a}^i - \boldsymbol{a}^j\right\|_2^2 s_{ij} + \lambda \sum_{q=1}^{3} h_q \left\|f_q(\boldsymbol{X}) - \boldsymbol{S}\right\|_F^2 \\ &\text{s.t. } \boldsymbol{S} \geq 0. \end{aligned} \tag{14}$$

Suppose that $\boldsymbol{G} = [g_{ij}]_{ij}$ with $g_{ij} = \|\boldsymbol{a}^i - \boldsymbol{a}^j$ Then, (14) can be converted to

$$\begin{aligned} &\min \gamma_2 \operatorname{Tr}\left(\boldsymbol{S}^T \boldsymbol{G}\right) + \lambda \sum_{q=1}^{3} h_q \left\|f_q(\boldsymbol{X}) - \boldsymbol{S}\right\|_F^2 \\ &\text{s.t. } \boldsymbol{S} \geq 0. \end{aligned} \tag{15}$$

Note that, with trace operator $\operatorname{Tr}(\cdot)$, $\|f_q(\boldsymbol{X}) - \boldsymbol{S}\|_F^2$ can be further expanded as

$$\begin{aligned} \left\|f_q(\boldsymbol{X}) - \boldsymbol{S}\right\|_F^2 &= \operatorname{Tr}\left(\left(f_q(\boldsymbol{X}) - \boldsymbol{S}\right)^T \left(f_q(\boldsymbol{X}) - \boldsymbol{S}\right)\right) \\ &= \operatorname{Tr}\left(f_q(\boldsymbol{X})^T f_q(\boldsymbol{X})\right) - 2\operatorname{Tr}\left(\boldsymbol{S}^T f_q(\boldsymbol{X})\right) \\ &\quad + \operatorname{Tr}\left(\boldsymbol{S}^T \boldsymbol{S}\right). \end{aligned} \tag{16}$$

By substituting (16) into (15) and removing the term $Tr(f_q(\mathbf{X})^T f_q(\mathbf{X}))$ irrelevant to $\boldsymbol{S}$, we can then obtain the equivalent form of (15) as

$$\begin{aligned} &\min \gamma_2 \operatorname{Tr}\left(\boldsymbol{S}^T \boldsymbol{G}\right) - 2\lambda \operatorname{Tr}\left(\boldsymbol{S}^T \left(\sum_{q=1}^{3} h_q f_q(\boldsymbol{X})\right)\right) \\ &\quad + \lambda \sum_q h_q \operatorname{Tr}\left(\boldsymbol{S}^T \boldsymbol{S}\right) \\ &\text{s.t. } \boldsymbol{S} \geq 0. \end{aligned} \tag{17}$$

Let $\boldsymbol{F} = \sum_{q=1}^{3} h_q f_q(\boldsymbol{X})$, then the objective function regarding $\boldsymbol{S}$ can be transformed as

$$\begin{aligned} &\min \gamma_2 \operatorname{Tr}\left(\boldsymbol{S}^T \boldsymbol{G}\right) - 2\lambda \operatorname{Tr}\left(\boldsymbol{S}^T \boldsymbol{F}\right) + \sum_{q=1}^{3} h_q \lambda \operatorname{Tr}\left(\boldsymbol{S}^T \boldsymbol{S}\right) \\ &\text{s.t. } \boldsymbol{S} \geq 0. \end{aligned} \tag{18}$$

After setting the derivative of (18) to zero, we can deduce the iterative formula of $\boldsymbol{S}$ as

$$\boldsymbol{S}^{(n+1)} = \left[\frac{2\lambda \boldsymbol{F}^{(n)} - \gamma_2 \boldsymbol{G}^{(n)}}{2\lambda \sum_{q=1}^{3} h_q{}^{(n)}}\right]_+ \tag{19}$$

where $[\cdot]_+$ is the indicator function for the nonnegative term. Take any variable z as an example, if z is larger than zero, then $[z]_+ = z$; otherwise, $[z]_+ = 0$. Therefore, we can acquire the updating formula of $\boldsymbol{S}$ based on (19).

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3) Fix $\boldsymbol{E}$ and $\boldsymbol{S}$ and Optimize $\boldsymbol{A}$: In view of calculating $\boldsymbol{A}$, it is evident that only the items related to $\boldsymbol{A}$ should be considered. Equation (20) defines the specific objective function relevant to $\boldsymbol{A}$ by

$$\begin{aligned}\min\ & \gamma_1\|\boldsymbol{A}\|_{2,1}+\alpha\mathrm{Tr}(\boldsymbol{A})+\gamma_2\sum_i\sum_j\left\|\boldsymbol{a}^i-\boldsymbol{a}^j\right\|_2^2 s_{ij}\\ &+\mu_1\|\boldsymbol{X}-\boldsymbol{X}\boldsymbol{A}-\boldsymbol{E}\|_F^2\\ \text{s.t. }& \boldsymbol{A}\geq 0.\end{aligned}\tag{20}$$

Suppose that $\boldsymbol{L}_s = \boldsymbol{D} - ((\boldsymbol{S}+\boldsymbol{S}^T)/2)$ and $D_{ii} = \sum_{j=1}^{d}((s_{ij}+s_{ji})/2)$; then, we have

$$\begin{aligned}\min\ & \gamma_1\|\boldsymbol{A}\|_{2,1}+\alpha\mathrm{Tr}(\boldsymbol{A})+\gamma_2\mathrm{Tr}\left(\boldsymbol{A}^T\boldsymbol{L}_s\boldsymbol{A}\right)\\ &+\mu_1\|\boldsymbol{X}-\boldsymbol{X}\boldsymbol{A}-\boldsymbol{E}\|_F^2\\ \text{s.t. }& \boldsymbol{A}\geq 0.\end{aligned}\tag{21}$$

Let $\boldsymbol{U}=\mathrm{diag}(u_1,u_2,\ldots,u_d)\in\mathbb{R}^{d\times d}$ be a diagonal matrix with

$$u_i=\frac{1}{2\left(\|\boldsymbol{a}_i\|_2^2+\varsigma\right)^{\frac{1}{2}}}\quad(i=1,2,3,\ldots,d)$$

which is employed. Note that, different from $\boldsymbol{a}^i$, $\boldsymbol{a}_i\in\mathbb{R}^{d\times 1}$ denotes the ith column of $\boldsymbol{A}$. ς is a factor, e.g., $1e^{-16}$, used to avoid zero denominator in u_i. Evidently, with the introduction of $\boldsymbol{U}$, we can simplify $\|\boldsymbol{A}\|_{2,1}$ to $\mathrm{Tr}(\mathbf{A^T U A})$. Then, (21) can be converted to

$$\begin{aligned}\min\ & \gamma_1\mathrm{Tr}\left(\mathbf{A^T U A}\right)+\alpha\mathrm{Tr}(\boldsymbol{A})+\gamma_2\mathrm{Tr}\left(\boldsymbol{A}^T\boldsymbol{L}_s\boldsymbol{A}\right)\\ &+\mu_1\|\boldsymbol{X}-\boldsymbol{X}\boldsymbol{A}-\boldsymbol{E}\|_F^2\\ \text{s.t. }& \boldsymbol{A}\geq 0.\end{aligned}\tag{22}$$

After setting the derivative of (22) to zero, we can obtain the updating rule of $\boldsymbol{A}$ as follows:

$$\boldsymbol{A}^{(n+1)}=\boldsymbol{J}^{-1}\left[\mu_1\boldsymbol{X}^T\left(\boldsymbol{X}-\boldsymbol{E}^{(n+1)}\right)-\alpha\boldsymbol{I}\right]_+\tag{23}$$

where $\boldsymbol{J}=(2\gamma_2\boldsymbol{L}_s^{(n+1)}+2\gamma_1\boldsymbol{U}^{(n)}+\mu_1\boldsymbol{X}^T\boldsymbol{X})\in\mathbb{R}^{d\times d}$. $\boldsymbol{U}^{(n)}$ depicts the nth iteration of $\boldsymbol{U}$ on basis of $\boldsymbol{A}^{(n)}$. Analogously, $\boldsymbol{L}_s^{(n+1)}$ represents the $(n+1)$th iteration of $\boldsymbol{L}_s$ based on $\boldsymbol{S}^{(n+1)}$.

Therefore, with the initialization of $\boldsymbol{E}$, $\boldsymbol{S}$, and $\boldsymbol{A}$, it is achievable to obtain the sequential estimations of them via alternative iteration. Then, $\boldsymbol{X}$ can be reconstructed via self-contained regression, i.e., $\boldsymbol{X}\approx\boldsymbol{X}\boldsymbol{A}$. Hence, $\boldsymbol{a}^i(i=1,2,3,\ldots,d)$ actually implies the contribution of the ith band to regress $\boldsymbol{X}$.

In terms of uncovering the significance of the ith band, all entries in the ith row of $\boldsymbol{A}$ should be exploited. Therefore, the l_2-norm of each row vector from $\boldsymbol{A}$ is computed and used to characterize the representativeness of the corresponding band. Assume that $\boldsymbol{w}=[w_1,w_2,\ldots,w_d]^T$ is denoted as the significance vector involving each band. In specific, $w_i=\|\boldsymbol{a}^i\|_2$ is utilized to measure the significance of the ith band. Then, based on $\boldsymbol{w}$, the band prioritization can be determined. As a result, bands with the b greatest significance are selected to form the ultimate band subset.

Algorithm 1 summarizes the main procedure of the proposed unsupervised band selection method MGRSR.

Algorithm 1 Band Selection Process of MGRSR

Input: Original HSI $\boldsymbol{X}$, γ_1, γ_2, α, λ, μ_1, and tolerance parameter τ.

Output: band subset with b selected bands.

1: **Initialization:** $n=0$, $\boldsymbol{E}^n$, $\boldsymbol{S}^n$, $\boldsymbol{A}^n$
2: **repeat**
3: Update $\boldsymbol{E}^{n+1}$: according to (13);
4: Update $\boldsymbol{S}^{n+1}$: according to (19);
5: Update $\boldsymbol{A}^{n+1}$: according to (23);
6: **until** $|\boldsymbol{A}^{(n+1)}-\boldsymbol{A}^{(n)}|<\tau$, $|\boldsymbol{S}^{(n+1)}-\boldsymbol{S}^{(n)}|<\tau$, and $|\boldsymbol{E}^{(n+1)}-\boldsymbol{E}^{(n)}|<\tau$.
7: Calculate band importance vector $\boldsymbol{w}$ on basis of $w_i=\|\boldsymbol{a}^i\|_2(i=1,2,3,\ldots,d)$.
8: Sort $\boldsymbol{w}$ in descending order.
9: **return** b selected bands with respect to the b largest $\boldsymbol{w}$.

IV. Experiments

In this section, to evaluate the performance of MGRSR, comparative experiments are conducted on different hyperspectral data sets. First, we introduce the experimental setups, involving the hyperspectral data sets, comparison methods, classification algorithms, and parameter analysis in Section IV-A. Second, Sections IV-B, IV-C, and IV-D provide the performance comparison of MGRSR with seven representative unsupervised band selection methods on three hyperspectral data sets, respectively. For the sake of assessing the performance of MGRSR comprehensively, we also analyze the classification results with all bands. Third, in Section IV-E, to further investigate the effectiveness of multigraph term in MGRSR, the ablation experiments are given with different single or double graphs. Then, we analyze the robustness of the bands selected by MGRSR with respect to different proportions of hyperspectral data in Section IV-F. After that, Section IV-G analyzes the convergence of MGRSR. Finally, the computational time of ten band selection methods are compared in Section IV-H.

A. Experimental Setup

We adopt three widely used benchmark hyperspectral data sets, i.e., Houston, Pavia University, and Indian Pines for comparative experiments.

1) Hyperspectral Data Set: a) The Houston hyperspectral data set was captured by the Compact Airborne Spectrographic Imager (CASI) over the University of Houston campus and the neighboring urban area. It has 349×1905 pixels with a spatial resolution of 2.5 m, containing 144 spectral bands ranging from 380 to 1050 nm. There are 15 labeled classes in this image.

b) The Pavia University data set was collected by the Reflective Optics System Imaging Spectrometer (ROSIS)-3 over the Pavia University. The spatial resolution is about 1.3 m with the size of 610×340. In this image, there are nine labeled classes, i.e., Asphalt, Meadows, Gravel, Trees, Painted metal sheets, Bare Soil, Bitumen, Self-Blocking Bricks, and Shadows.

c) The Indian Pines data set was acquired by the Airborne Visible/Infrared Spectrometer (AVIRIS) over

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TABLE II
RANGE OF GRID SEARCH FOR PARAMETERS

Parameters	range of grid search	Indian Pines	Pavia University	Houston
α	1, 3, 6, 12, 20	6	6	6
γ_1	1, 10, 15, 20, 25, 30, 32	25	32	30
λ	10, 40, 80, 100, 140	100	100	100
γ_2	1, 5, 10, 15, 20, 25, 30, 32	32	25	15
μ_1	1, 10, 20, 40, 80	80	40	80

Northwestern Indiana in 1992. It has 16 classes with 145×145 pixels. The spatial resolution is about 3.7 m. In this image, there are 224 spectral reflectance bands ranging from 0.4 to 2.5 μm. Due to the removal of 20 water absorption bands, i.e., [104–108], [150–163], and 220, only 204 bands are reserved and employed in the experiments.

2) Comparison Methods: To evaluate the effectiveness of the proposed band selection method, i.e., MGRSR, seven representative unsupervised band selection methods include: 1) clustering-based E-FDPC [34], WaLuDi [33], and TRC-OC-FDPC [35]; 2) sparsity-based Laplacian score method (LSC) [65] and $l_{2,1}$-norm regularized discriminative method (UDFS) [66]; 3) searching-based VGBS [39]; and 4) graph embedding-based manifold-preserving and weakly redundant method (MPWR) [31]. Moreover, to comprehensively assess the performance of MGRSR, this article also analyzes the classification results with all bands.

In the comparison methods, all the parameters are provided by the authors or determined via fivefold cross validation. For example, in terms of WaLuDi, bands are first quantized to 0–255. While, for MPWR, α is set as 0.1, 0.1, and 1, ρ is set as 5, 25, and 30 on Indian Pines, Pavia University, and Houston data sets, respectively. Simultaneously, μ is set as 8 on three data sets. In view of UDFS, the regularization parameter is set as 10^6.

3) Classification Algorithms: For the performance evaluation of MGRSR, four classification algorithms include the 3-D convolutional neural network method for hyperspectral image classification algorithm (3-D-CNN) [67], the support vector machine (SVM) [5], [68]–[70], the linear discriminant analysis (LDA) [71], and the edge-preserving filtering-based spectral–spatial classification algorithm (EPF) [72] as classifiers. 3-D-CNN, abbreviated as CNN in this article, is a typical deep spectral–spatial HSI classification [73]. SVM is an excellent classifier for small data set and has been widely used for band selection methods [37], [40]. LDA focuses on the separability of data and is often utilized to evaluate the effectiveness of various band selection methods. EPF has achieved superior classification performance due to the utilization of both spectral information and spatial information.

The patch size in CNN is set as 3×3, whereas the epochs are set as 20, 20, and 40 in Indian Pines, Pavia University, and Houston data sets, respectively. Regarding SVM, C and γ are set as 5000 and 0.1, respectively. In terms of EPF, the numbers for neighborhood k are set as 4, 9, and 13 with respect to Houston, Pavia University, and Indian Pines images.

Note that the validation set used to tune the parameters of the model is generally adopted in machine learning practice. Thus, in the experiments, around 8% and 2% data are randomly selected as the training and validation set, respectively, and the remaining 90% are used as the testing set.

Then, after a certain number of bands are selected with band selection methods, a compact HSI composed of the selected bands can be obtained. Then, by employing classification algorithms, the performance of selected bands can be evaluated. In this article, two commonly used classification performance evaluation criteria, i.e., overall accuracy (OA) and kappa statistic coefficient (κ), are employed to assess the effectiveness of the proposed methods.

4) Parameter Analysis: As illustrated in (6), α, γ_1, λ, γ_2, and μ_1 act as vital parameters. To investigate the impact of different parameters, relevant experiments are performed on the validation set. In the experiments, the grid search strategy is adopted. Specifically, to avoid too many parameters of the grid search, the ranges of α, γ_1, λ, γ_2, and μ_1 are listed in the first two columns of Table II.

Evidently, based on the classification results on the validation set, we can capture the optimal setting of α, γ_1, λ, γ_2, and μ_1, as tabulated in the last three columns of Table II. Note that the settings of k are consistent with EPF on three data sets.

As listed in Table II, the optimal settings of α and λ are the same on three data sets. Meanwhile, except for the Houston data set, the best values of μ_1 are consistent on both Indian Pines and Pavia University data sets. For intuitive display, Fig. 1 plots the sensitivity analysis of MGRSR to α, λ, and μ_1 on three data sets. In each panel, the x-axis represents the number of selected bands, whereas the y-axis denotes the OA values produced by SVM, LDA, and EPF under different settings of parameters. Note that, in each panel, irrelevant parameters are fixed according to Table II. For example, to depict the impact analysis of α on MGRSR on basis of the Houston data set, γ_1, λ, γ_2, and μ_1 are set as 30, 100, 15, and 80, respectively.

From Fig. 1, we can observe that, on both Pavia University and Houston validation sets, the best value of α corresponds to 6, whereas the optimal one of λ is 100 under three classifiers. Despite that EPF exhibits weak performance fluctuation on the Indian Pines validation set, for α and λ, both LDA and SVM give the same optimal settings as on the other two validation sets. Regarding μ_1, the best value is set as 80 on both Indian Pines and Pavia University data sets. This indicates that MGRSR is not sensitive to α, λ, and μ_1.

Analogously, Figs. 2 and 3 illustrate the influence curves of γ_1 and γ_2 on MGRSR. Apparently, due to the excellent performance, regardless of the values of γ_1 and γ_2, EPF always achieves high classification accuracy, while, for SVM and

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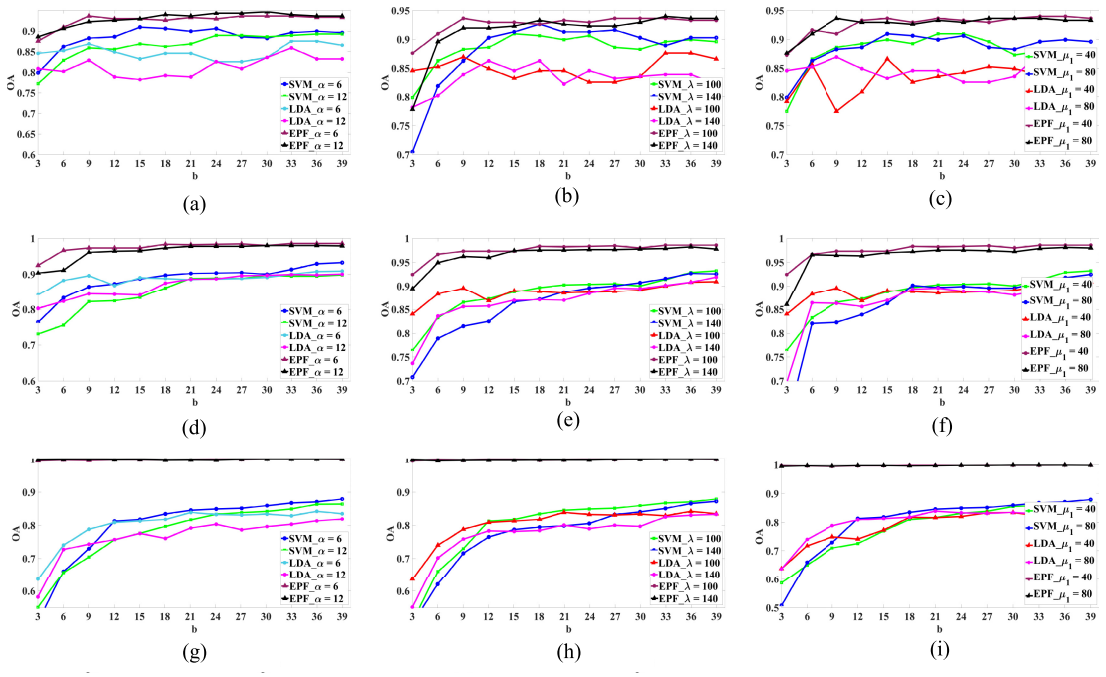


Fig. 1. Curves of classification accuracy versus the number of selected bands under different choices of α, λ, and μ_1 on three data sets. (a) Houston image. (b) Houston image. (c) Houston image. (d) Pavia university image. (e) Pavia university image. (f) Pavia university image. (g) Indian pines image. (h) Indian pines image. (i) Indian pines image.

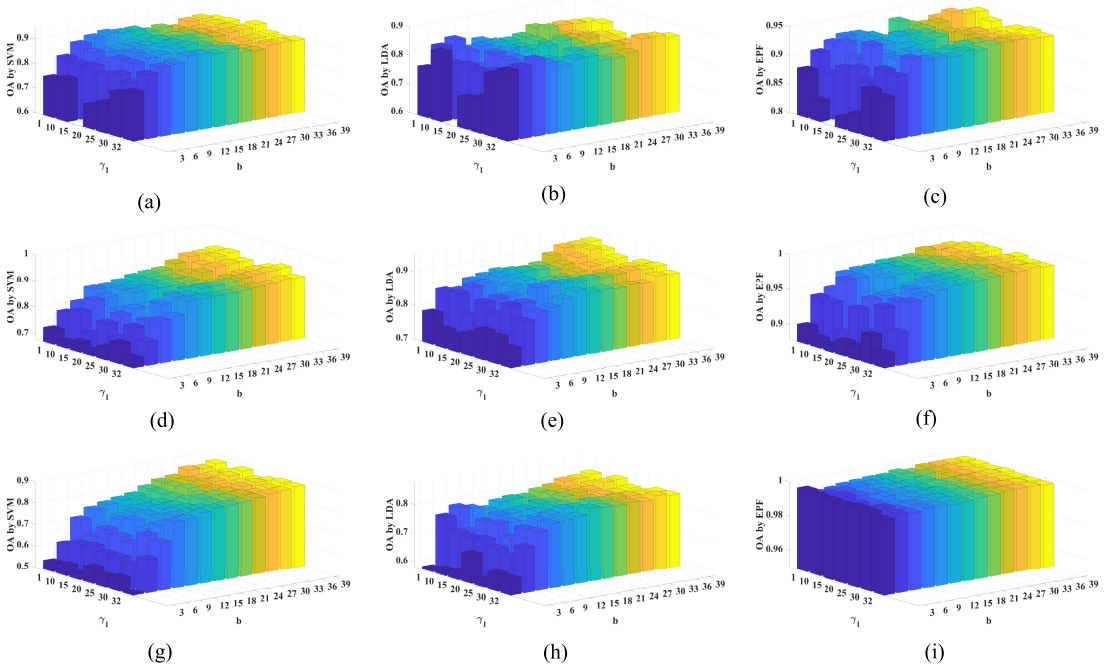


Fig. 2. Curves of classification accuracy versus the number of selected bands under different choices of γ_1 on three data sets. Panels in each column represent the average OA values produced by SVM, LDA, and EPF. (a) Houston image. (b) Houston image. (c) Houston image. (d) Pavia University image. (e) Pavia University image. (f) Pavia University image. (g) Indian Pines image. (h) Indian Pines image. (i) Indian Pines image.

LDA, the change of γ_1 and γ_2 will lead to the floating of classification accuracy.

B. Experimental Results on Houston Data Set

We further evaluate the performance of MGRSR on the Houston data set as done to the other two data sets. Fig. 4 plots the curves of classification accuracies varying with b for ten methods.

From Fig. 4, we can see that MGRSR performs well in terms of CNN-, SVM-, LDA-, and EPF-based classification accuracies. For instance, as shown in Fig. 4(b)–(d), WaLuDi, TRC-OC-FDPC, MPWR, and MGRSR produce superior per-

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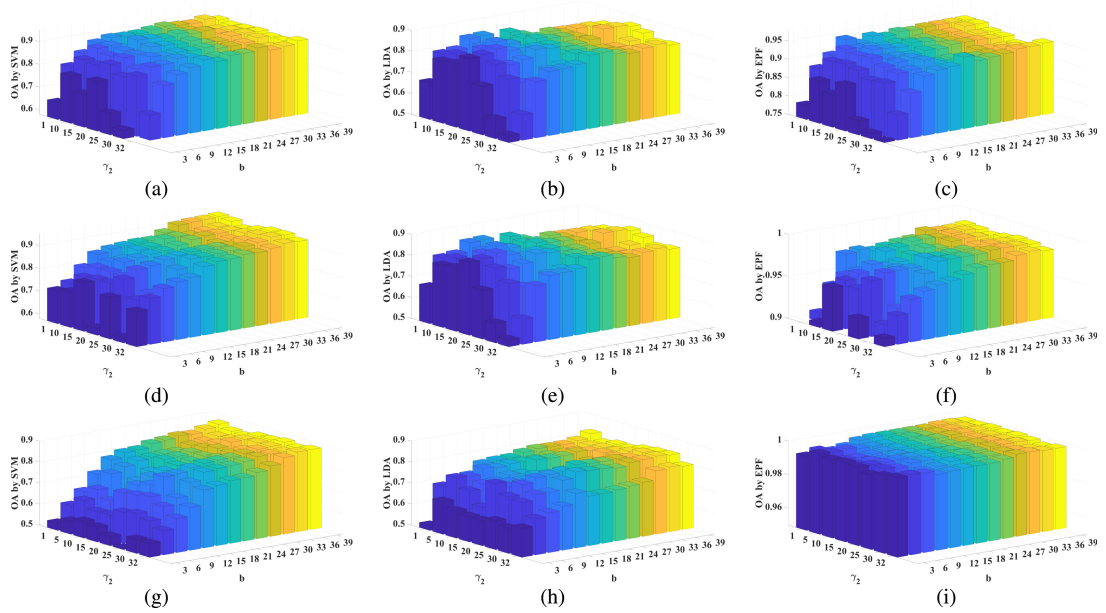


Fig. 3. Curves of classification accuracy versus the number of selected bands under different choices of γ_2 on three data sets. Panels in each column represent the average OA values produced by SVM, LDA, and EPF. (a) Houston image. (b) Houston image. (c) Houston image. (d) Pavia university image. (e) Pavia university image. (f) Pavia university image. (g) Indian pines image. (h) Indian pines image. (i) Indian pines image.

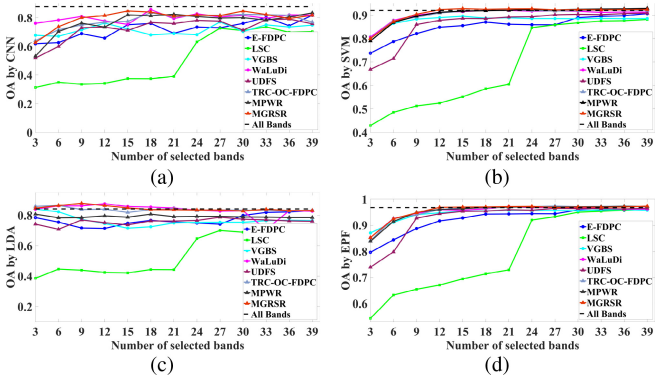


Fig. 4. OA of different band selection methods on the Houston data set with different numbers of selected bands: (a)–(d) CNN-, SVM-, LDA-, and EPF-based OA curves, respectively. The black dotted lines mean that all bands are used for classification.

formance. Particularly, they can achieve comparable and even higher classification accuracies with all bands for b greater than 12.

From Fig. 4(a), we can observe that the performance of different methods varies greatly. In most cases, MGRSR, MPWR, and WaLuDi have an evident performance advantage over the others. Take the accuracy of b equal to 12 as an example. The LDA-based OA values for MGRSR and WaLuDi are 0.8639 and 0.8756, respectively, while the lowest one produced by others is 0.4240. Since LDA mainly depends on the separability of features, the classification results reflect the distinctiveness differences of selected bands. Therefore, the effectiveness of MGRSR can be manifested.

The main reason for the robust and excellent performance of MGRSR is that the integrated interband multigraph makes the derived regression coefficient matrix not only uncovers band-wise importance values but also coherent to the comprehensive interband relationship. With the coefficient matrix, we can acquire reasonable and generic band importance measures; in turn, the effectiveness and generalization of the MGRSR can be improved. In addition, due to the introduction of joint trace minimum and nonnegative constraints, the trivial solution can be avoided, and the reliability of the method is further enhanced.

To give a quantitative comparison, Table III describes the CNN-, SVM-, LDA-, and EPF-based OA and κ of each method with b being 12.

C. Experimental Results on Pavia University Data Set

To assess the effectiveness of MGRSR, the performances of ten band selection methods are compared based on the classification accuracies produced by CNN, SVM, LDA, and EPF. Meanwhile, the classification accuracies with all bands are also compared. For illustrative purposes, Fig. 5 shows the curves of classification accuracies versus the number of selected bands b under ten band selection methods and all bands. In this figure, the x-axis represents the number of selected bands, whereas the y-axis characterizes the classification accuracy denoted by OA.

As depicted in Fig. 5, as b increases, the classification accuracy rises accordingly. Specifically, for b less than 18, the increase in b is usually accompanied by obvious improvement of OA. The reason for this is that, when b is low, more bands are required to supplement more discriminative and complementary information for classification.

From Fig. 5, we can see that most methods perform well. In particular, for b greater than 12, the accuracy curves of VGBS, TRC-OC-FDPC, MPWR, and MGRSR tend to be stable and very close, while, for b less than 9, MGRSR is superior to most methods, as shown in Fig. 5(c) and (d). For example, when b changes from 3 to 6, the LDA- and

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TABLE III

OA (IN %) AND κ STATISTIC OBTAINED BY CNN, SVM, LDA, AND EPF ON THE HOUSTON DATA SET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD TYPEFACE

Methods	E-FDPC	LSC	VGBS	WaLuDi	UDFS	TRC-OC-FDPC	MPWR	MGRSR
Classifier					CNN			
OA	65.82	34.31	76.26	77.76	73.42	76.90	73.44	**81.55**
κ	63.01	29.23	74.17	75.79	71.11	75.05	71.13	**79.93**
Classifier					SVM			
OA	84.83	52.55	88.94	91.18	87.78	90.92	91.22	**92.31**
κ	83.58	48.91	88.02	90.45	86.76	90.16	90.49	**91.67**
Classifier					LDA			
OA	71.30	42.40	74.51	**87.56**	75.06	84.03	79.55	86.39
κ	68.96	37.93	72.42	**86.52**	72.98	82.69	77.85	85.25
Classifier					EPF			
OA	91.63	67.04	94.77	95.99	93.46	96.02	96.06	**96.87**
κ	90.94	64.52	94.34	95.66	93.92	95.69	95.73	**96.61**

TABLE IV

OA (IN %) AND κ STATISTIC OBTAINED BY CNN, SVM, LDA, AND EPF ON THE PAVIA UNIVERSITY DATA SET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD TYPEFACE

Methods	E-FDPC	LSC	VGBS	WaLuDi	UDFS	TRC-OC-FDPC	MPWR	MGRSR
Classifier					CNN			
OA	90.25	84.90	96.53	95.95	91.07	93.73	96.47	**96.62**
κ	87.09	80.36	95.44	94.78	88.05	91.66	95.36	**95.57**
Classifier					SVM			
OA	79.31	58.37	**86.78**	85.94	78.38	84.99	85.96	86.44
κ	71.28	37.74	**82.15**	80.91	69.77	79.54	80.90	81.62
Classifier					LDA			
OA	80.35	77.66	**90.06**	87.55	86.23	86.01	86.23	87.73
κ	73.15	69.31	86.68	83.20	81.36	81.02	81.36	**83.42**
Classifier					EPF			
OA	83.04	78.72	**91.91**	90.42	86.76	89.34	91.12	91.30
κ	76.48	70.10	**89.09**	87.00	81.83	85.45	88.18	88.22

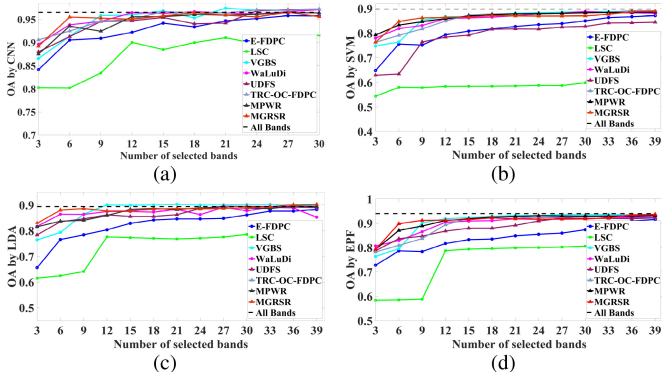


Fig. 5. OA of different band selection methods on the Pavia University data set with different numbers of selected bands: (a)–(d) CNN-, SVM-, LDA-, and EPF-based OA curves, respectively. The black dotted lines mean that all bands are used for classification.

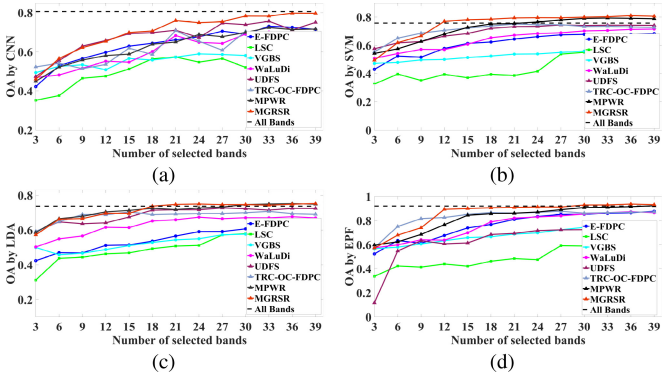


Fig. 6. OA of different band selection methods on the Indian Pines data set with different numbers of selected bands: (a)–(d) CNN-, SVM-, LDA-, and EPF-based OA curves, respectively. The black dotted lines mean that all bands are used for classification.

EPF-based OA values of MGRSR increase by around 0.1. Although not always the best, MGRSR still exhibits overall satisfying classification performance. These experimental results show the robustness and superiority of MGRSR.

In addition, to give the quantitative comparison with respect to different methods, Table IV tabulates the OA and κ of each method for $b = 12$.

D. Experimental Results on Indian Pines Data Set

To evaluate the effectiveness of MGRSR, we present a performance comparison of MGRSR with seven representative methods on the Indian Pines data set by CNN, SVM, LDA, and EPF classifiers. Notably, concerning holistic performance evaluation, the classification accuracies with all bands are also compared. For illustrative purpose, Fig. 6 plots the curves of CNN-, SVM-, LDA-, and EPF-based classification accuracies versus the number of selected bands b under ten band selection methods and all bands. In this figure, the x-axis represents the number of selected bands, whereas the y-axis characterizes the classification accuracy denoted by OA.

From Fig. 6(a), we can find that, for b less than 12, MPWR has pretty good performance. When b locates between 12 and 30, MGRSR is obviously superior to the others. Notably, when b is 18, the OA value obtained by MGRSR

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TABLE V

OA (IN %) AND κ STATISTIC OBTAINED BY CNN, SVM, LDA, AND EPF ON THE INDIAN PINES DATA SET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD TYPEFACE

Methods	E-FDPC	LSC	VGBS	WaLuDi	UDFS	TRC-OC-FDPC	MPWR	MGRSR
Classifier					CNN			
OA	59.76	47.55	50.87	55.16	**65.74**	53.59	58.06	65.33
κ	53.80	38.83	44.15	48.16	**60.81**	45.99	51.81	60.51
Classifier					SVM			
OA	57.83	39.44	49.98	56.97	57.29	70.66	58.73	**77.03**
κ	50.97	28.26	41.97	49.53	50.08	66.23	51.94	**73.66**
Classifier					LDA			
OA	51.20	46.27	48.74	61.60	64.19	68.87	70.43	**69.63**
κ	42.28	36.38	39.99	55.53	58.55	63.76	65.87	**64.86**
Classifier					EPF			
OA	67.59	43.98	63.33	63.79	60.71	82.65	76.59	**89.36**
κ	61.71	31.67	56.12	57.07	53.42	80.02	72.83	**87.77**

is over 0.1 higher than the suboptimal one. Moreover, for b being 39, MGRSR can achieve the OA value close to that of all bands.

Fig. 6(b) and (d) manifests that, for b less than 12, TRC-OC-FDPC usually outperforms the others. In comparison, when b is greater than 15, MPWR also exhibits excellent performance. Despite the superior performance of TRC-OC-FDPC and MPWR, MGRSR has an obvious advantage over them for b greater than 15. Specifically, when b is between 18 and 30, MGRSR can yield higher OA than all bands without exception.

As demonstrated in Fig. 6(c), both TRC-OC-FDPC and MPWR perform well for b lower than 18, while, for b greater than 18, the performance of TRC-OC-FDPC degrades instead. One possible reason is that, as b increases, irrelevant and noisy bands are selected, resulting in performance deterioration. Specifically, analogous to the abovementioned comparison results, MGRSR outperforms the others for b greater than 18. It is worth noting that, at this time, MGRSR can even achieve better performance than all bands. Therefore, the robustness and effectiveness of MGRSR are further revealed.

To give the quantitative comparison with respect to different methods, Table V lists the OA and κ of each method for $b = 12$. Specifically, for illustrative purposes, we take the EPF-based classification maps as an example. The classification maps corresponding to the abovementioned band selection methods for b being 12 are depicted in Fig. 7.

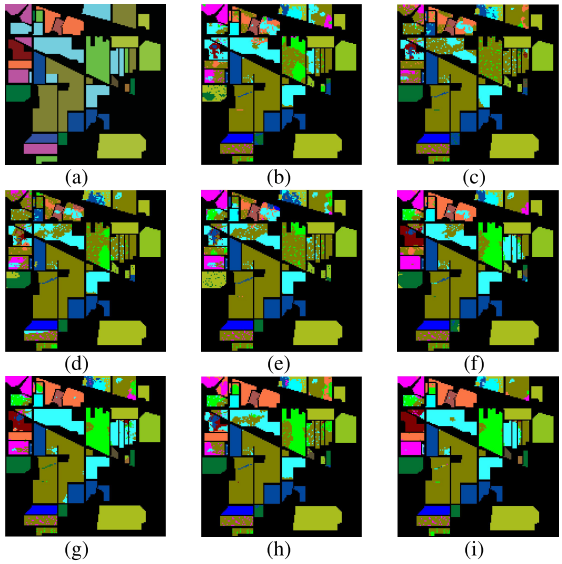


Fig. 7. EPF-based classification maps on the Indian Pines data set with 12 bands selected by (a) ground truth, (b) E-FDPC, (c) LSC, (d) VGBS, (e) WaLuDi, (f) UDFS, (g) TRC-OC-FDPC, (h) TRC-OC-FDPC, (i) MPWR, and (j) MGRSR.

E. Effectiveness Analysis of Multigraph Term

To investigate the effectiveness of multigraph term in MGRSR, experiments with different single or double graphs on three data sets are conducted. Figs. 8 and 9 describe the corresponding experimental results on Indian Pines and Pavia University data sets, respectively. In each figure, concerning the case of employing single graph, $\boldsymbol{f_1}$, $\boldsymbol{f_2}$, and $\boldsymbol{f_3}$ represents that only $f_1(\boldsymbol{X})$, $f_2(\boldsymbol{X})$, or $f_3(\boldsymbol{X})$ is employed, respectively. Meanwhile, $\boldsymbol{f_12}$, $\boldsymbol{f_13}$, and $\boldsymbol{f_23}$ correspond to three pairs of double graphs. For instance, $\boldsymbol{f_12}$ denotes the case that $f_1(\boldsymbol{X})$ and $f_2(\boldsymbol{X})$ are utilized.

As depicted in Figs. 8 and 9, as the number of selected bands increases, compared with different single or double graphs, MGRSR can perform well in most cases. The main reason for this is that multigraph helps to better make up the shortcomings of diverse graphs. In this regard, the reasonability and generalization of the interband neighborhood graph can be improved. Consequently, MGRSR is capable of gaining representative bands favorable to classification.

F. Robustness to Data Size

In the above section, the experimental results have demonstrated the effectiveness and robustness of MGRSR on different data sets. To further evaluate the robustness of MGRSR to data size, we adopt four percentages, i.e., 10%, 40%, 80%, and 100%, of the data to select 39 bands on three data sets for ten times. In addition, as TRC-OC-FDPC and MPWR perform well in the above experiments, they are utilized as comparison methods. Fig. 10 depicts the 3-D stem diagrams for the experimental results.

In the figures, the x-axis represents the band number, and the y-axis is the percentage of data, while the z-axis reveals the frequency that the corresponding band is selected in ten independent experiments. Evidently, the frequency ranges from 1 to 10. From Fig. 10(a), (d), and (g), we can find that,

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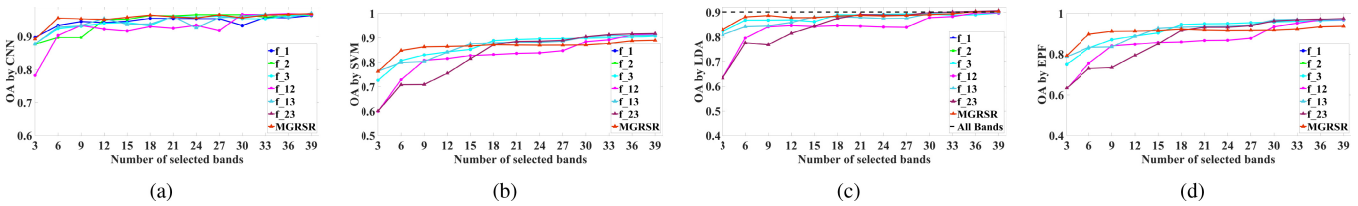


Fig. 8. Performance comparison of multigraph with single and two graphs on the Pavia University data set: (a)–(d) CNN-, SVM-, LDA-, and EPF-based OA curves, respectively. The black dotted lines mean that all bands are used for classification.

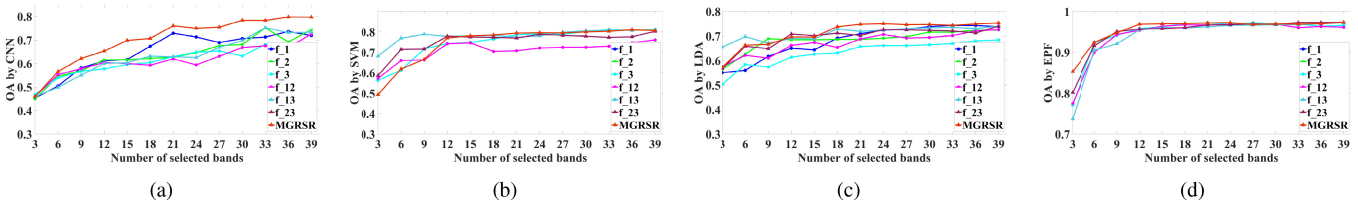


Fig. 9. Performance comparison of multigraph with single and two graphs on the Indian Pines data set: (a)–(d) CNN-, SVM-, LDA-, and EPF-based OA curves, respectively. The black dotted lines mean that all bands are used for classification.

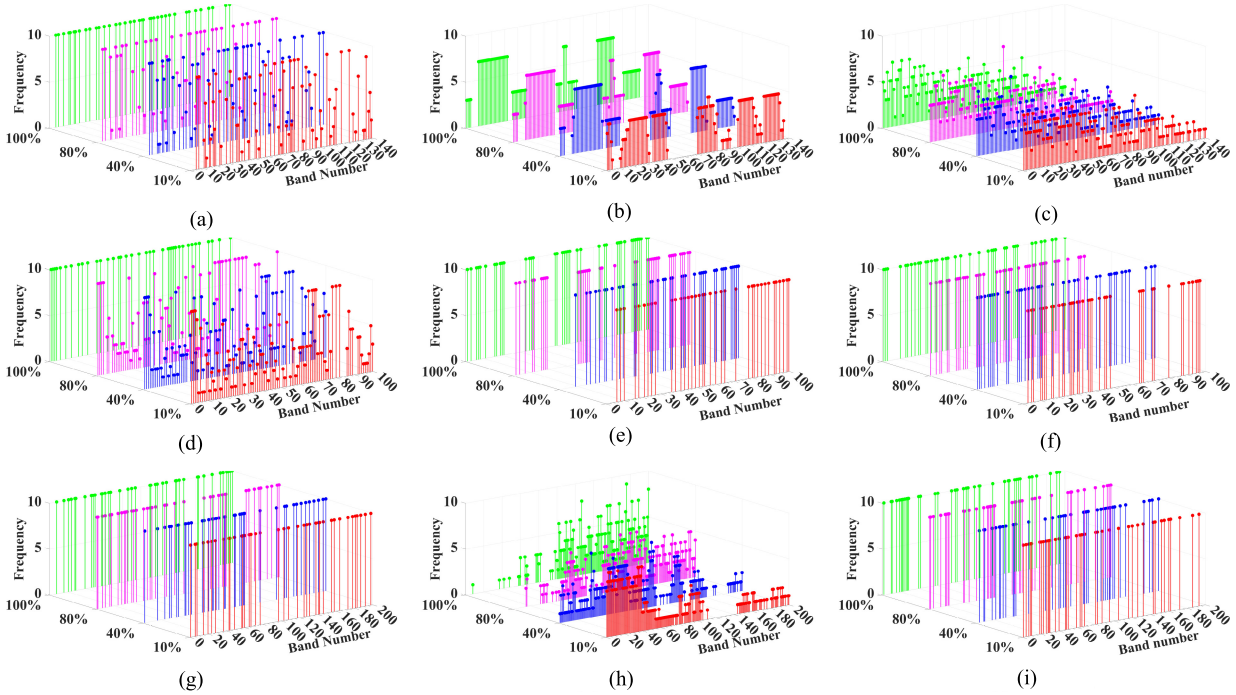


Fig. 10. 40 selected bands based on four percentages of data, i.e., 10%, 40%, 80%, and 100%, for ten times on three data sets: (a)–(c), (d)–(f), and (g)–(i) experimental results on Houston, Paiva University, and Indian Pines data sets. The z-axis denotes the number of times that the band was selected in ten experiments. (a) TRC-OC-FDPC. (b) MPWR. (c) MGRSR. (d) TRC-OC-FDPC. (e) MPWR. (f) MGRSR. (g) TRC-OC-FDPC. (h) MPWR. (i) MGRSR.

with 10%, 40%, or 80% proportions of data, the selected frequencies of each band by TRC-OC-FDPC are similar, while, when 100% data are utilized, the results of band selection are quite different from the previous three cases. In terms of MPWR, it is capable of making close choices with four data sizes on the Houston and Pavia University data sets. However, on the Indian Pines data set, the selected bands vary with four proportions of data. Compared with TRC-OC-FDPC and MPWR, the bands selected by MGRSR are more consistent on three data sets based on four sizes of data. This indicates that MGRSR is robust to data size as well.

G. Convergence Analysis

Note that convergence is quite important to the efficiency and reliability of an algorithm. In this connection, this section focuses on analyzing the convergence of MGRSR. When the change of $\boldsymbol{U}$ is ignored, (22) is convex with respect to $\boldsymbol{A}$. The reason for this is that, after fixing $\boldsymbol{E}$ and $\boldsymbol{S}$, the objective function related to $\boldsymbol{A}$ involves typical convex functions. Then, it will converge to a global minimum by alternating updating $\boldsymbol{E}$, $\boldsymbol{A}$, and $\boldsymbol{S}$. Analogously, since (18) shows that the objective value related to $\boldsymbol{S}$ is convex, it can also converge to a global minimum. Note that the objective function

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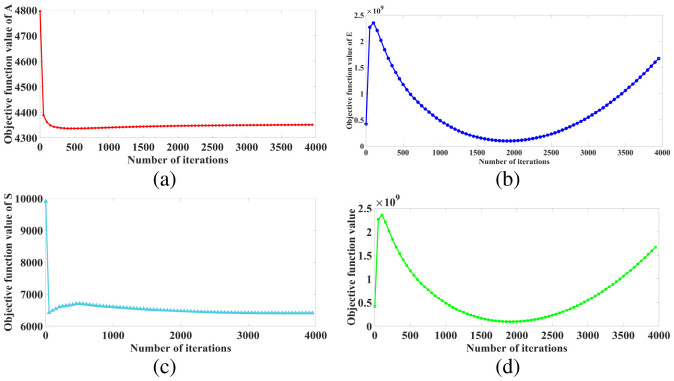


Fig. 11. Convergence of MGRSR: (a)–(d) plot the curves of the objective function values w.r.t. A, E, S , and overall objective function versus the number of iterations respectively.

TABLE VI

RUNNING TIMES (s) OF EACH BAND SELECTION METHOD. THE LONGEST TIME IS HIGHLIGHTED IN BOLD TYPEFACE

Method	12	20	30	Time Complexity
E-FDPC	0.88	0.86	0.83	$\mathcal{O}\left(Nd^2\right)$
LSC	0.49	0.49	0.49	$\mathcal{O}\left(N^2d\right)$
VGBS	0.58	0.62	0.60	$\mathcal{O}\left(\sum_{i=b+1}^{d} i^3 + d^3\right)$
WaLuDi	0.56	0.56	0.56	$\mathcal{O}\left(Nd^3\right)$
UDFS	19.27	19.27	19.27	$\mathcal{O}\left(d^3 + dN^2\right)$
TRC-OC-FDPC	0.13	0.19	0.23	$\mathcal{O}\left(d^3 + Nd^2\right)$
MPWR	**47.33**	**46.47**	**46.77**	$\mathcal{O}\left(n\left(d^3 + N^2d\right)\right)$
MGRSR	15.67	15.67	15.67	$\mathcal{O}\left(n\left(d^3 + d^2N\right)\right)$

about $\boldsymbol{E}$ is nonconvex, only the local minimum can be captured. Then, by combing the three situations, we can infer that the proposed method will converge to a local minimum value.

In this article, we take the Houston data as an example and plot the convergence curves, as shown in Fig. 11. In this figure, the x-axis denotes the number of iterations, while the y-axis represents the objective function values of $\boldsymbol{E}$, $\boldsymbol{A}$, $\boldsymbol{S}$, and the overall objective function values. From Fig. 11(b), we can see that there is a sharp increase. This is because after adopting random initializations of both **A** and **E**, and the residuals $\|\boldsymbol{X} - \boldsymbol{X}\boldsymbol{A} - \boldsymbol{E}\|_F^2$ are large. In this connection, the value of the overall objective function exhibits a sharp increase as well.

H. Computational Complexity

This section mainly analyzes the computational complexity of MGRSR. It should be noted that, in most cases, $N > d$. Then, during each iteration of MGRSR, the cost for updating $\boldsymbol{E}$ is $\mathcal{O}(Nd^2)$. As calculating $\boldsymbol{G}$ and $\boldsymbol{F}$ needs $\mathcal{O}(d^3)$ and $\mathcal{O}(Nd^2)$, respectively, the cost for updating $\boldsymbol{S}$ is $\mathcal{O}(Nd^2 + d^3)$. In terms of updating $\boldsymbol{A}$, the computational costs corresponding to $\boldsymbol{J}$, $\boldsymbol{J}^{-1}$, and $\boldsymbol{J}^{-1}\boldsymbol{X}^T(\boldsymbol{X} - \boldsymbol{E}^{(n+1)})$ are $\mathcal{O}(Nd^2)$, $\mathcal{O}(d^3)$, and $\mathcal{O}(Nd^2)$. Hence, updating $\boldsymbol{A}$ requires $\mathcal{O}(d^3 + Nd^2)$. Therefore, the total cost for MGRSR is $\mathcal{O}(t(d^3 + d^2N))$. In comparison, the computational costs of the other seven methods are listed in Table VI.

To further compare the run time of various band selection methods, comparative experiments are conducted on a computer with Intel Core i7-9700 3.00-GHz CPU and 16-GB RAM. First, three data sets are all downsampled based on the window of 100 × 100. Then, all methods are implemented in MATLAB R2017a. Table VI provides the running times of each method for b equal to 12, 20, and 30, respectively. In this table, the execution times are measured in seconds. Moreover, considering the statistical significance of experimental results, all the running times are averaged over twenty independent runs.

Table VI demonstrates that E-FDPC and VGBS have high computational efficiency. For MGRSR, it costs less time than UDFS and more time than LSC, while, for $N \gg d$ and $N \gg t$, LSC will be more time-consuming than MGRSR based on the time complexity.

V. CONCLUSION AND DISCUSSION

This article presents a robust unsupervised band selection method with MGRSR. In this method, the integrated interband neighborhood graph is merged into the sparse self-contained regression model. This is advantageous to gain a more reliable and reasonable bandwise significance measure, which conforms to the interband structure instead of arbitrary. Benefiting from the combination of multiple interband graphs from different perspectives, it is favorable to integrate the strengths of multiple graphs to realize complementarity and then enhance the robustness of the model. Moreover, the introduction of joint trace minimum and nonnegative constraint contributes to improving the reliability of bandwise significance. Besides, to solve the problem, a corresponding iterative update algorithm is devised.

Comparative experiments with several classical band selection methods are conducted on Indian Pines, Pavia University, and Houston data sets. Throughout the experiments, we can find that MGRSR often performs well and is robust on different data sets. Since γ_2 and γ_1 have an evident influence on the performance of MGRSR, the choice of γ_2 and γ_1 is of great value.

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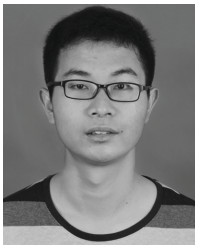
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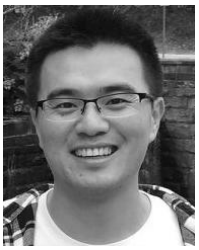
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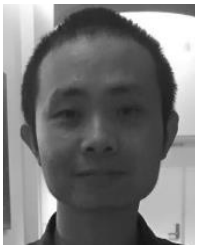
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